

Lecture 11 - Economics and Business Administration
Financial Ratios

Agenda

- Liquidity Ratios
- Solvency Ratios
- Profitability Ratios
- Financial Leverage
- Working Capital Management

Overview

- Financial ratios **complement reclassification** by **summarizing and** quantifying complex phenomena
- They support the analysis of three dimensions:
  - **Profitability**: relative to correlated metrics
  - **Financial Solvency**: long-term solvency of the firm
  - **Liquidity**: short-term solvency of the firm
- **Ratio analysis does not fully address all informational needs for evaluating economic performance**
  - *Internal and external facts and elements provide additional context*
- Ratio analysis becomes more meaningful when **performed over multiple periods** and compared to **industry averages**



Liquidity Ratios

Liquidity and Solvency Ratios: Balance Sheet — Current/Non-Current Classification

<table>
<tr><th>ASSETS</th><th>LIABILITIES</th></tr>
<tr><td>SHORT-TERM ASSETS
(convertible into cash within 12 months)</td><td rowspan="2">SHORT-TERM LIABILITIES
(due within 12 months)</td></tr>
<tr><td rowspan="3">MEDIUM-LONG TERM ASSETS
(convertible into cash after 12 months)</td></tr>
<tr><td>MEDIUM-LONG TERM LIABILITIES
(due after 12 months)</td></tr>
<tr><td>EQUITY</td></tr>
</table>

Liquidity Ratios

- Ability of the company to meet short-term payment obligations
- Balance Sheet classified using the Current/Non-Current classification (IAS 1)

Name	Formula	Benchmark
Net Liquid Assets	$Immediate\ \&\ deferred\ liquidity - Current\ liabilities$	>0
Quick Ratio	$(Current\ assets - Inventories) / Current\ liabilities$	>1
Net Working Capital (NWC)	$Current\ assets - Current\ liabilities$	>0
Current Ratio	$Current\ assets / Current\ liabilities$	≈2

Liquidity Ratios

- **Net Liquid Assets**: if > 0, there is **a cash surplus**
- **Quick Ratio**:
  - Provides a **more reliable assessment** of liquidity ➔ Excludes Inventories!
- **Current Ratio**:
  - **As the ratio increases, the probability grows** that **short-term obligations** can be met **through** liquid assets **without additional** borrowing
- **NWC:** highlights the company's short-term financial balance or imbalance
  - NWC > 0: **Financial Equilibrium**
  - NWC < 0: **Financial Imbalance** → The company will collect less in the short term than needed to honor its maturing liabilities

Liquidity Ratios: Reflections

- **NWC > 0**
- **Net Liquid Assets < 0**

Difficulty in meeting **short-term** financial obligations: the company's most illiquid current assets (typically inventory) **cannot be fully converted into cash.**

- **Low predictive value**
- Must be considered in relation to dynamic interactions with solvency, profitability, and growth
- Derived from Balance Sheet values: they are **static** ➔ Easily distorted (e.g. accelerating/delaying collections from customers or payments to suppliers, …)



Solvency Ratios

Solvency Ratios

- Indicators of **financial solidity**: long-term solvency (*degree of dependence on third-party financing*)

Name	Formula	Benchmark
Debt-to-equity ratio	$\frac{Total\ Liabilities\ (third\ party\ funds)}{Equity\ (own\ funds)}$	≈1
Fixed Assets Coverage Ratio	$\frac{Equity\ (own\ funds)}{Net\ Technical\ Fixed\ Assets\ (tangible\ +\ intangible)}$	>1
Structural Margin	$Equity - Net\ Technical\ Fixed\ Assets$	>0

Solvency Ratios

- **Debt-to-Equity Ratio** → grade **of dependence on third-party financing**, expresses the ratio of Total Debt to Equity
  - = 1: Balanced position
  - > 1: Liabilities exceed equity; leverage exceeds shareholders' contribution
- **Fixed Asset Coverage Ratio** → relates equity to long-term assets instrumental to core operations
  - **> 1: Solid company** → Fixed assets financed exclusively with equity
  - < 1: Physiological in certain business life stages (e.g., start-up phase; high-growth phases with significant investments)
- **Structural Margin** →same variables as the Fixed Asset Coverage Ratio



Profitability Ratios

Capital Employed Approach

IS

Core Operations
Non-Core Activities
Financial Management
Exceptional Items (Non-recurring)
Tax Management
Net Income

BS

<table>
<tr><th>ASSETS</th><th>LIABILITIES</th></tr>
<tr><td rowspan="2">Current Operating Assets</td><td>Current Operating Liabilities</td></tr>
<tr><td rowspan="2">Financial Liabilities</td></tr>
<tr><td>Operating Non-Current Assets</td></tr>
<tr><td>Non-Core Assets</td><td>Net Tax Position</td></tr>
<tr><td colspan="2">Exceptional (Non-recurring) Assets / Liabilities</td></tr>
<tr><td>Cash and Cash Equivalents</td><td>Equity</td></tr>
</table>

Profitability Ratios

- **Profitability**: a relative concept
  - *A measure of income is related to a correlated base figure*
- Profitability **ratios** measure the **company's ability** to **generate income** and adequately **remunerate** all **factors of production**
- Main profitability ratios:
  - **Return on Investment (ROI)**: return on invested capital
  - **Return on Equity (ROE)**: return on shareholders' equity

Profitability Ratios: Income Statement — Margin Format

Net Sales
- Variable Costs
= Contribution Margin
- Fixed Operating Costs
= EBITDA
- Depreciation and Amortization
= EBIT
+/- Financial Income (Expenses)
= Earnings Before Taxes
- Taxes
= Net Income



Earnings Before Interest, Taxes, Depreciation, Amortization

Measures the company's ability to generate margins from core operations, excluding depreciation, amortization, and financial items. Useful for comparing operating performance across **companies with different capital structures.**



Earnings Before Interest and Taxes

Indicates operating income net of depreciation and amortization, highlighting **the impact of investment policies**. A key indicator for assessing the long-term economic sustainability of the core business.

Profitability Ratios: ROI — Return on Investment

$$ROI = \frac{EBIT}{\text{Capital Employed (Net Invested Capital)}}$$

It confronts **economic result from core and ancillary activities** (before income taxes, exceptional (non-recurring) items, and financial expenses) to the **capital employed** to produce that income ➔ showcases the **return on all company investments**

Profitability Ratios: ROI Decomposition

$$ROI = \frac{EBIT}{CE}$$

$$= \frac{CE\ core}{EBIT\ core} * \frac{CE\ core}{CE\ tot} + \frac{EBIT\ anc}{CE\ anc} * \frac{CE\ anc}{CEtot}$$

Profitability Ratios: ROA – Return on Assets

$$ROA = \frac{EBIT}{Total\ Assets}$$

ROA = ROS × AT

ROS
= EBIT / Net Sales

AT
= Net Sales / Total Assets

What ROA measures

- ROA expresses operating income as a percentage of total assets — the return generated per euro of assets invested, **regardless of how they are financed**.

Two drivers

1. **Return on Sales (ROS)**: margin earned per euro of revenue. Driven by pricing power (luxury, pharma) or cost discipline (online banking, retail)
2. **Asset Turnover (AT):** how efficiently assets generate revenue.
   High AT → high-volume low-margin model (retail). Low AT high margin model (e.g. custom-built yacht manufacturing)

ROA vs. ROI

Both use EBIT in the numerator. The denominator differs:

- *ROA = EBIT / Total Assets (as-reported BS: includes all assets)*
- *ROI = EBIT / Capital Employed (reclassified BS: excludes extraordinary and financial assets)*

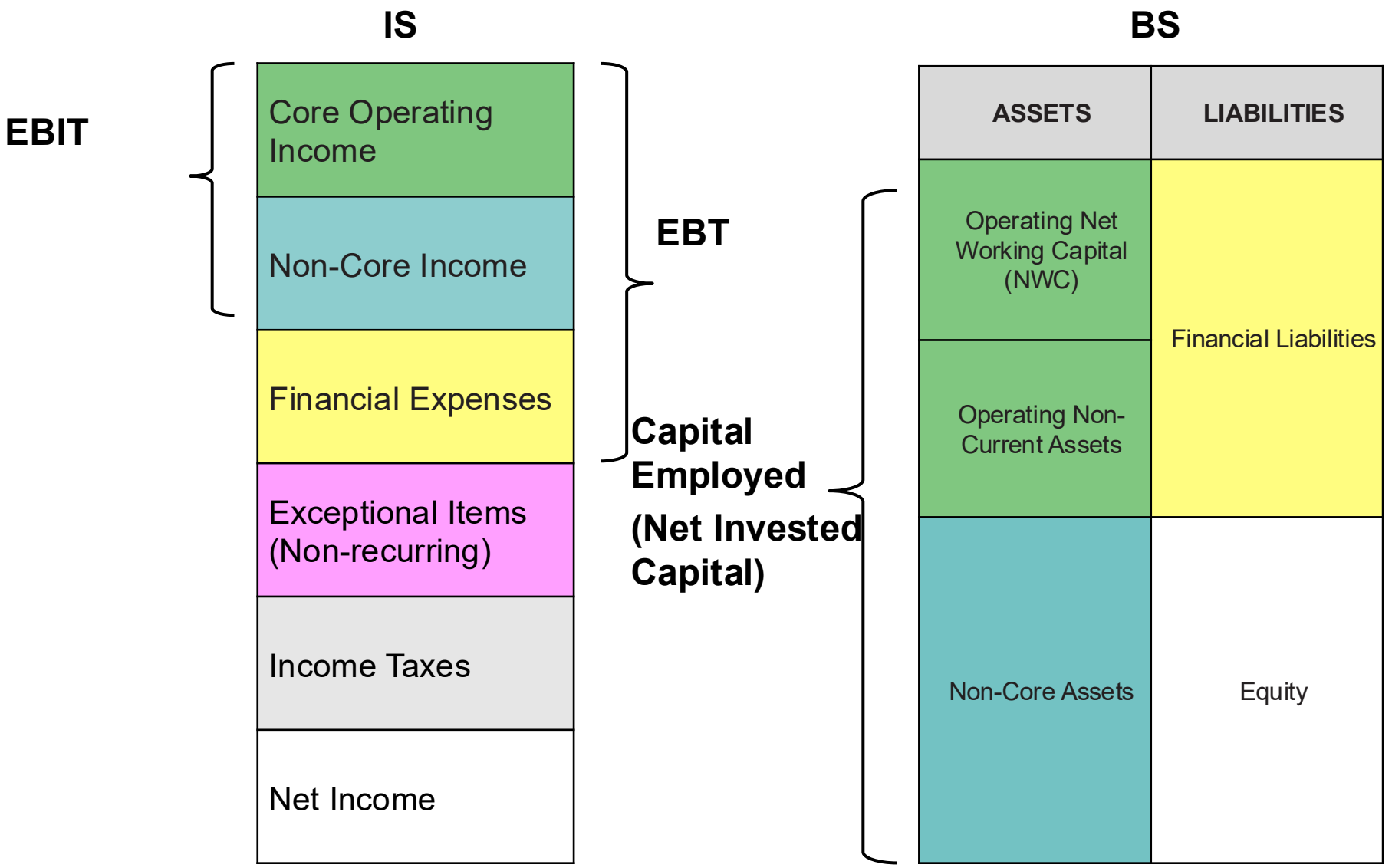
- ROI is analytically richer: it isolates operational performance
- ROA is easier to compute from public data and useful for sector benchmarking

Profitability Ratios: ROE

$$ROE = \frac{\text{NET INCOME}}{\text{EQUITY}}$$

- **Return** on **equity** during the period, **before** any **decision** to **distribute dividends** → — the return on the investment made by shareholders in the company
  - *Potential increase in equity during the period*
  - *Rate of return for risk capital providers*

Profitability Ratios: ROE Decomposition



Profitability Ratios: ROE Decomposition

- $\boldsymbol{RONA}\ (Return\ on\ Net\ Asset) = \dfrac{EBIT}{Net\ Operating\ Capital\ Invested}$

- **D** = Financial Debt (D)
- **Cost of financial liabilities** = i = Financial Interest / Debt
- **NI** = Net Income
- **EBT** = Earnings Before Taxes (EBT) before exceptional items and income taxes

$$ROE = \left[RONA + (RONA - i) * \frac{D}{E}\right] * \frac{NI}{EBT}$$

Profitability Ratios: ROE Decomposition

The ability of the company to remunerate risk capital (ROE) depends on:

- **RONA**: return on Net Assets, ability to generate operating income relative to net operating capital invested
- **Spread** between **operating profitability (RONA)** and **the cost of financial liabilities** (RONA– i)
- **Financial Leverage Ratio**: level of financial debt relative to equity $\frac{D}{E}$
- **Tax & Exceptional Items Incidence Rate**: impact of taxes and exceptional items on income $\frac{NI}{EBT}$

$$ROE = \left[RONA + (\boldsymbol{RONA} - i) * \frac{D}{\boldsymbol{E}}\right] * \frac{NI}{EBT}$$

ROE Decomposition: Financial Leverage

- If $RONA > \frac{Interest}{Debt}$ ➔ multiplier effect on the return on invested capital (ROE) through the debt ratio $\frac{D}{E}$
- The more indebted the company, the > will be the ROE

- Third-party capital works for equity holders and as long as the cost of borrowed funds is lower than the return obtainable by investing those funds in the company's operations.
- In practice, **as debt levels rise, the associated cost also increases**, **since higher leverage means greater perceived risk for lenders**
- **An increase in debt can improve profitability (ROE)** if borrowed funds cost less than the return they generate in terms of RONA.
- However**, a progressive increase in debt carries the risk of raising the cost of borrowed capital to the point where the profitability generated by core operations is partially or fully offset by financial expenses**

Financial Leverage: Example

$$ROE = \left[RONA + (RONA - i) * \frac{D}{E}\right] * \frac{NI}{EBT}$$

$$15\% = [15\% + (15\% - 5\%) * 1] * 0{,}6$$

If Alfa were able to increase its debt from 1 to 3 with no change in any other component of the formula, what would the effect be?

Financial Leverage: Example

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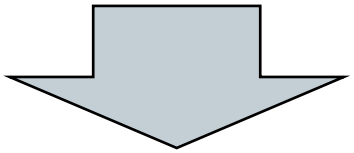
$$[15\% + (15\% - 5\%) * 3] * 0{,}6 = 27\%$$

The return for shareholders would increase from 15% to 27%

Financial Leverage Analysis: Market Perspective

ROE = {ROI + D/E * (ROI - r)}

- The financial leverage formula shows how operating and financial management interact in determining the company's overall profitability (ROI, cost of debt (i), D/E).
- The relationship between operating profitability (ROI) and the return for shareholders (ROE) depends on the relationship between ROI and i
- If ROI > i, ROE > ROI: overall profitability increases the more debt capital is used

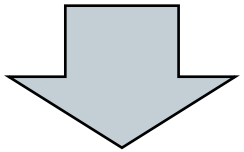


For companies able to earn a return on invested capital greater than the cost of debt: IS IT ALWAYS BENEFICIAL TO INCREASE EXTERNAL FINANCING INDEFINITELY?

Financial Leverage Analysis: Market Perspective

It is NOT advisable to increase debt (D/E) beyond a certain threshold), this is why:

- as the level of debt increases, **the average cost demanded by lenders is likely to rise** (**reflecting the higher business risk they perceive)**
- **it must also be kept in mind that ROI is a stochastic variable,** not known with certainty in advance



Equal deviations from forecasts in operating profitability (ROI) or in the cost of debt are amplified the higher the leverage ratio (D/E); financial leverage increases the variability of results, that is, it raises risk and consequently the cost of capital.



Working Capital Management

Duration Ratios

The size of Operating NWC is closely linked to the following financial ratios:

Days Sales Outstanding (DSO): trade receivables / daily sales

$$\frac{\text{Trade receivables net of VAT}}{\text{Sales Revenue / 360}}$$

Days Payable Outstanding (DPO): debts /daily purchases

$$\frac{\text{Trades payables net of VAT}}{\text{Purchases of goods and services / 360}}$$

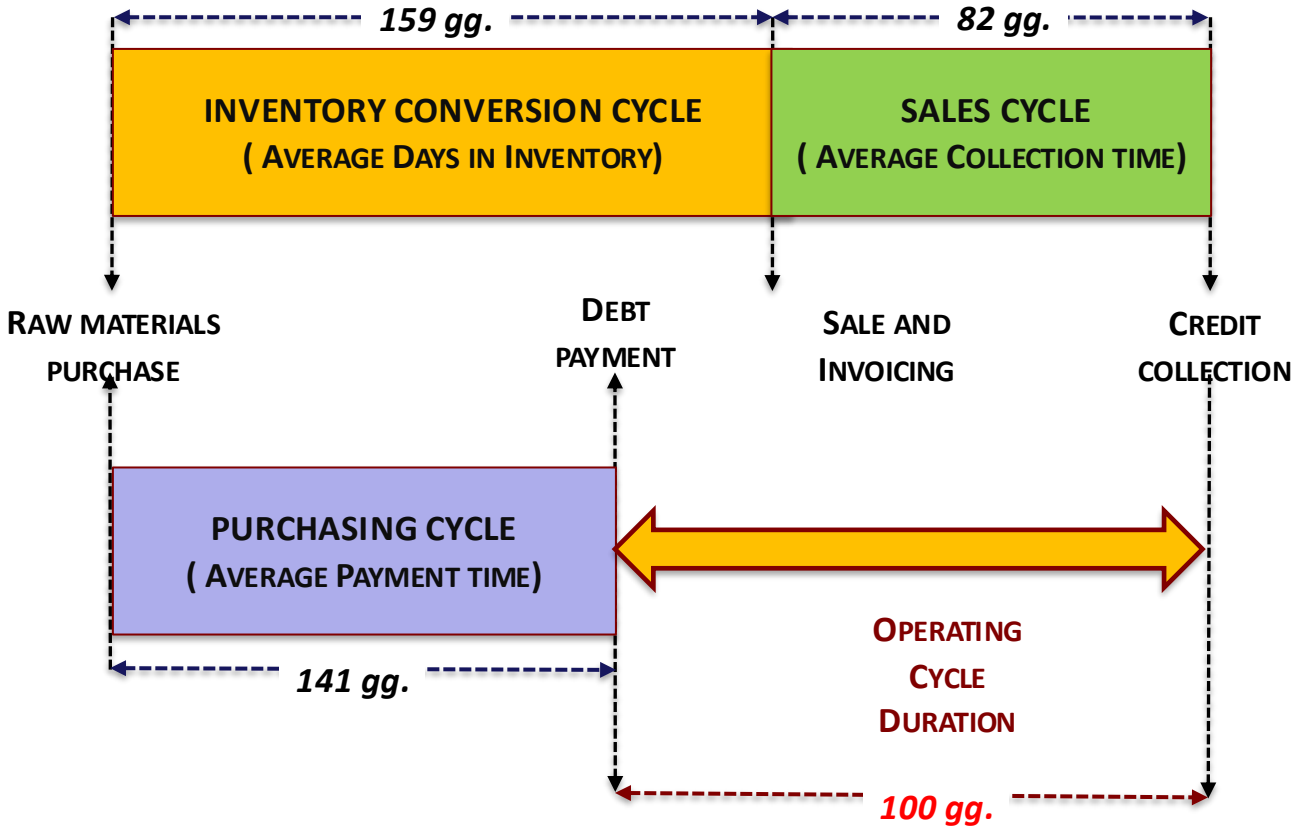
Days Inventory Outstanding (DIO) : inventory/ daily production cost

$$\frac{\text{Inventory}}{\text{Cost of Goods Sold / 360}}$$

Duration Ratios

Working capital needs to be financed!

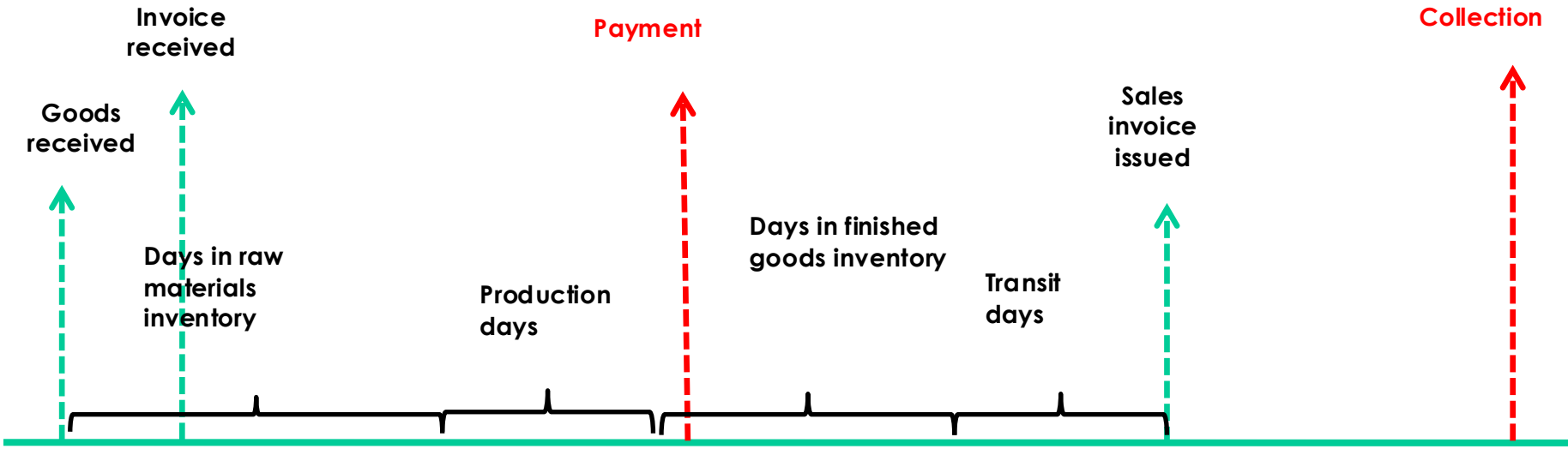




In practice, a widely used ratio measures operating net working capital relative to net sales:

Financial need % = Operating NWC / Net Sales

The Working Capital Cycle



Storage and production period

- Optimize safety stock levels
- Optimize production planning policies
- Integrate sales forecast information
- Monitor slow-moving inventory
- Improve logistics flows

Payment period

- Optimize early payment discounts
- Assign payment responsibilities to business functions
- Put suppliers in competition at equal price/quality/service ratio

Collection period

- Monitor payment terms compliance
- Improve collection process
- Control order management process to avoid anomalies and delays
- Optimize invoicing timing
- Optimize timing by VAT recovery on bad debts



Campari Group – Case Study

Financial Indicators – Campari Group:

Item	Value
Net Sales	
EBIT (Operating result"	
Net Income (group share)	
Financial Expenses (gross)	
Financial Income	
Net Financial Expenses	
D&A	
EBITDA (calculated)	
Total Assets	
Total Equity	
Bonds (non-current)	
Loans due to banks (non-current)	
Other non-current fin. liab.	
Loans due to banks (current)	
Other current fin. liab.	
Financial Debt (gross)	
Cash & equivalents	
Net Financial Debt	

Financial Indicators – Campari Group:

Item	Value
Net Sales	€3,051.2m
EBIT (Operating result)	€567.5m
Net Income (group share)	€333.1m
Financial Expenses (gross)	€120.3m
Financial Income	€69.6m
Net Financial Expenses	€50.7m
D&A	€148.3m
EBITDA (calculated)	€715.8m
Total Assets	€8,111.4m
Total Equity	€3,864.2m
Bonds (non-current)	€1,590.1m
Loans due to banks (non-current)	€627.6m
Other non-current fin. liab.	€138.8m
Loans due to banks (current)	€272.2m
Other current fin. liab.	€69.7m
Financial Debt (gross)	**€2,698.4m**
Cash & equivalents	€703.3m
Net Financial Debt	**€1,958.0m**

Financial Ratios – Campari Group

Ratio	Formula	Calculation	Result
ROS			
AT			
ROA			
CE			
RONA = ROI			
ROE			
D/E			
i			
Spread			
Interest coverage			

Financial Ratios – Campari Group

Ratio	Formula	Calculation	Result
ROS	EBIT / Net Sales	567.5 / 3,051.2	**18.6%**
AT	Net Sales / Total Assets	3,051.2 / 8,111.4	**0.38×**
ROA	EBIT / Total Assets	567.5 / 8,111.4	**7.0%**
CE	Equity + NFD	3,864.2 + 1,958.0	**€5,822.2m**
RONA = ROI	EBIT / CE	567.5 / 5,822.2	**9.7%**
ROE	Net Income / Equity	333.1 / 3,864.2	**8.6%**
D/E	NFD / Equity	1,958.0 / 3,864.2	**0.51×**
i	Net Fin. Exp. / NFD	50.7 / 1,958.0	**2.6%**
Spread	RONA − i	9.7% − 2.6%	**+7.2 pp**
Interest coverage	EBIT / Fin. Exp. (gross)	567.5 / 120.3	**4.7×**

Financial Indicators – Campari Group: Profitability

- **ROS = 18.6%** → High by sector standards: reflects strong pricing p**ower of premium brands** (Aperol, Campari, Grand Marnier). Margin is supported by a favourable product mix and advertising spend used as a strategic lever rather than a cost drag
- **Asset Turnover = 0.38×** → Very low: typical of companies with an asset base inflated by **acquired intangibles** (goodwill €2,233m + brands €1,144m = 42% of total assets). This is not operational inefficiency: it is the accounting consequence of a **growth-by-acquisition strategy**.
- **ROA = 7.0%** →In line with the spirits/luxury sector average

➢ the result of the ROS/AT trade-off (ROA = ROS × AT = 18.6% × 0.38).

- **ROI = 9.7%** → Slightly above ROA because Capital Employed excludes part of non-operating assets.
  Solid but not outstanding: signals that the acquisitive growth strategy has diluted returns on capital compared to more asset-light operators.
- **ROE = 8.6%** → Relatively modest despite positive leverage: partly due **to non-recurring items** compressing net income in 2025 (impairments, earn-out remeasurements). Normalised structural ROE would be materially higher.

Financial Indicators – Campari Group: Leverage & Solvency

- **Financial Debt (gross) = €2,698m**
- **NFD = €1,958m**
  Gross debt is significant but a strong cash position (€703m) brings net debt down to €1,958m. The predominantly **long-term structure** (bonds + non-current loans) limits short-term refinancing risk.
- **D/E = 0.51×** → Conservative leverage: for every euro of equity, the group carries 51 cents of net debt. Campari historically operated at higher leverage; the current de-leveraging reflects financial discipline following the Courvoisier acquisition (2024).
- **i = 2.6%** → Very low net cost of debt_ benefits from fixed-rate bonds issued in favourable rate environments and a solid liquidity position.
- **Spread RONA − i = +7.2 pp** → Wide positive spread: operating profitability comfortably exceeds the cost of third-party capital. This means every additional euro of debt creates value for shareholders ➔ financial leverage is working in favour of ROE.
- **Interest coverage = 4.7×** → EBIT covers gross financial expenses nearly 5 times, a comfortable level, coverage improves significantly when excluding non-recurring financial items (earn-out changes, hyperinflation effects)